

Project 7

Signed IK Classification of PUMA 560 and UR5

Format: Group of two students; simulation-based project.

Prerequisites: D–H modelling, analytical inverse kinematics, Jacobians, determinant factorization, and singularity classification.

Motivation

A useful IK label describes a geometric configuration and the singular boundary separating it from alternatives. PUMA 560 and UR5 offer two different 6R architectures for testing this idea. Comparing them shows how determinant factorization can support an interpretable classification without assuming that identical joint numbers imply identical geometry.

Task

Overall objectives. Compare the PUMA and UR5 architectures, derive and interpret the three factors of the UR5 Jacobian determinant, associate regular IK solutions with signed categories, and deduce whether UR5 is cuspidal or noncuspidal.

- **Compare the two kinematic models.** Inspect the URDFs, STL geometry, axis arrangements, and D–H conventions. Identify the PUMA spherical wrist and the different UR5 wrist/parallel-axis structure. Validate both forward models and document joint offsets, base frames, and tool frames.
- **Derive and check the UR5 determinant.** Construct the full geometric Jacobian and factor its determinant into three configuration-dependent components, ignoring nonzero constant factors. Verify the factorization numerically against direct Jacobian evaluation and interpret each zero set geometrically.
- **Assign signed categories.** Use the provided IK routines to obtain eight regular mathematical solutions at a suitable target for each robot. Assign $(\text{sgn } f_1, \text{sgn } f_2, \text{sgn } f_3)$ to every UR5 solution and compare with PUMA’s shoulder–elbow–wrist labels. Check the one-to-one association at additional regular targets and separate joint-limit filtering from mathematical classification.
- **Visualize the separators.** Plot UR5 determinant factors on selected (q_2, q_3) slices at explicitly fixed q_4, q_5 and other joint values. Relate these slices to full six-dimensional configurations without treating projected off-slice IKs as on-slice points. For the PUMA arm, also map reduced singularities and the two-/four-arm-IK workspace distribution.
- **Deduce cuspidality from structure.** Explain why changing a factor sign along a continuous path requires a zero crossing. Combine this fact with the complete regular IK classification to infer UR5’s cuspidality status; factor count or determinant sign alone is not a sufficient argument. Test small approaches to each separator and report ambiguous near-zero labels.

Specifics and scope

Study two architectures, three regular targets per robot, and one short approach to each UR5 singularity type. The UR5 URDF, STL meshes, and IK implementation will be provided, together with a PUMA IK baseline. Focus new derivation on the UR5 Jacobian and classification rather than implementing another general IK solver. Allow geometric sign conventions to differ between models.

Check both translation and orientation residuals for every claimed full-pose IK. Document angle periodicity and limits, Jacobian conventions, and any scaling used for singular values. State all fixed coordinates in reduced plots; a two-dimensional slice does not show the entire 6R singular set. Refine decisive transitions, and distinguish sampled evidence from a continuous-path guarantee. Collision freedom is not implied by these kinematic checks.

Expected outcome

Explain both robot geometries and show how each UR5 IK fits into a signed category of three determinant factors. Deliver the verified factorization, illustrated classification tables, singularity plots, and a justified deduction of whether UR5 is cuspidal or noncuspidal.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Provided materials and implementation

The PUMA 560 and UR5 robot descriptions (.urdf) and link meshes (.stl) will be provided to the students, together with the analytical IK routines for both robots. Use a robotics library or your own tools for axis/frame inspection, IK visualization, and singularity plots. Confirm joint offsets and the selected terminal frame when importing each model. Use the supplied UR5 version consistently instead of substituting another UR-series robot or dimensional variant.

Supporting reading: *Kinematic issues in 6R cuspidal robots*, especially the discussion of determinant factors and UR-type architectures. Explain the argument in your own model conventions rather than copying a factorization without checking it.

Validate the supplied IK outputs against your forward models and use the full Jacobians to check the derived factors and reduced visualizations.